

Algebra B - T2 Cayley–Hamilton and Its Consequences

1. Matrix Powers via Cayley–Hamilton

Proposition 1.1 (Reduction of Powers)

Let $A \in M_2(\mathbb{R})$. Then for every $n \geq 2$, there exist $a, b \in \mathbb{R}$ such that

$$A^n = aA + bI.$$

Proof

Let $\chi_A(X)$ be the characteristic polynomial of A . By polynomial division,

$$X^n = q(X)\chi_A(X) + r(X), \quad \deg(r) < 2.$$

Thus,

$$r(X) = aX + b.$$

Evaluating at A and using Cayley–Hamilton,

$$A^n = aA + bI.$$

Computation of Coefficients

Assume 3 is a root of χ_A . Then

Evaluate at $X = 3$:

$$3^n = 3a + b$$

Differentiate:

$$nX^{n-1} = q'(X)\chi_A(X) + q(X)\chi_A'(X) + a$$

Evaluate at $X = 3$:

$$a = n3^{n-1}$$

Substitute back:

$$b = 3^n - 3(n3^{n-1}) = (1 - n)3^n$$

Conclusion

$$A^n = n3^{n-1}A + (1-n)3^n I$$

Remark

$$A - 3I = \begin{pmatrix} 0 & -1 \\ 0 & -1 \end{pmatrix}$$

has rank 1, hence the geometric multiplicity is strictly less than the algebraic multiplicity. Therefore,

A is not diagonalizable.

2. Inverse as a Polynomial in A

Proposition 2.1

Let $A \in M_n(\mathbb{R})$ be invertible. Then A^{-1} is a linear combination of $I, A, \dots, A^{n-1}$.

Proof

Let

$$\chi_A(X) = X^n + a_{n-1}X^{n-1} + \dots + a_1X + a_0.$$

By Cayley–Hamilton,

$$A^n + a_{n-1}A^{n-1} + \dots + a_1A + a_0I = 0.$$

Rewriting,

$$A^n = -a_{n-1}A^{n-1} - \dots - a_1A - a_0I.$$

Since A is invertible,

$$a_0 = (-1)^n \det(A) \neq 0.$$

Multiply by A^{-1} :

$$A^{n-1} + a_{n-1}A^{n-2} + \dots + a_1I + a_0A^{-1} = 0.$$

Solving,

$$A^{-1} = -\frac{1}{a_0} A^{n-1} - \frac{a_{n-1}}{a_0} A^{n-2} - \dots - \frac{a_1}{a_0} I.$$

3. Cyclic Vector for Nilpotent Operators

Proposition 3.1

Let $A \in M_n(\mathbb{C})$ satisfy

$$A^n = 0, \quad A^{n-1} \neq 0.$$

Then there exists $v \in \mathbb{C}^n$ such that

$$v, Av, \dots, A^{n-1}v$$

is a basis of $\mathbb{C}^n$.

Proof

Since $A^{n-1} \neq 0$, there exists v such that

$$A^{n-1}v \neq 0.$$

Suppose

$$a_0v + a_1Av + \dots + a_{n-1}A^{n-1}v = 0.$$

Apply A^{n-1} :

$$a_0A^{n-1}v = 0 \Rightarrow a_0 = 0.$$

Apply A^{n-2} :

$$a_1A^{n-1}v = 0 \Rightarrow a_1 = 0.$$

Continuing,

$$a_0 = a_1 = \dots = a_{n-1} = 0.$$

Thus the set is linearly independent, and since it has n elements, it forms a basis.

4. Commuting Operators are Polynomials in A

Proposition 4.1

Let $A \in M_n(\mathbb{C})$ satisfy

$$A^n = 0, \quad A^{n-1} \neq 0,$$

and let $B \in M_n(\mathbb{C})$ satisfy

$$AB = BA.$$

Then there exists $p \in \mathbb{C}[X]$ such that

$$B = p(A).$$

Proof

Let $v, Av, \dots, A^{n-1}v$ be a basis as above.

Write

$$Bv = a_0v + a_1Av + \dots + a_{n-1}A^{n-1}v.$$

Define

$$p(X) = a_0 + a_1X + \dots + a_{n-1}X^{n-1}.$$

For $0 \leq i \leq n-1$,

$$B(A^i v) = A^i(Bv).$$

Thus,

$$B(A^i v) = \sum_{j=0}^{n-1} a_j A^{i+j} v.$$

On the other hand,

$$p(A)(A^i v) = \sum_{j=0}^{n-1} a_j A^{i+j} v.$$

Hence,

$$B(A^i v) = p(A)(A^i v).$$

Since they agree on a basis,

$$B = p(A).$$

5. Final Summary

- Cayley–Hamilton reduces all powers of A to degree $< n$
- If A is invertible, then A^{-1} is a polynomial in A
- Maximal nilpotency implies existence of a cyclic basis
- Any operator commuting with such A is a polynomial in A